

W02

MATH 135

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Contents

1	Q1	2
1.1	(a)	2
1.2	(b)	3
2	Problem 2	5
3	Problem 3	10
4	Q4	12
5	Q5	14
6	Q6	15

1 Q1

1.1 (a)

Problem 1. Let K , L , and M be statement variables and suppose that the following logical expression is TRUE:

$$\neg((L \vee (\neg M)) \implies (\neg K)) \wedge (\neg L).$$

Determine with justification all possible truth value combinations of K , L , and M .

Solution. According to the extra lecture, we have already proved that

$$(A \implies B) \equiv \neg A \vee B.$$

Therefore, we have

$$((L \vee (\neg M)) \implies (\neg K)) \equiv (\neg(L \vee (\neg M)) \vee (\neg K)).$$

Then, by De Morgan's law, we get

$$(\neg(L \vee (\neg M)) \vee (\neg K)) \equiv ((\neg L \wedge M) \vee (\neg K)).$$

Thus, we have

$$\neg((\neg L \wedge M) \vee (\neg K)) \wedge (\neg L).$$

Applying De Morgan's law again, we get

$$\neg((\neg L \wedge M) \vee (\neg K)) \wedge (\neg L) \equiv (\neg(\neg L \wedge M) \wedge \neg(\neg K)) \wedge (\neg L).$$

Applying De Morgan's law once more, we have

$$(\neg(\neg L \wedge M) \wedge \neg(\neg K)) \wedge (\neg L) \equiv ((L \vee \neg M) \wedge K) \wedge (\neg L).$$

By the associative law, this is equivalent to

$$(L \vee \neg M) \wedge K \wedge (\neg L).$$

As a final step, we construct a truth table:

L	M	K	$\neg L$	$L \vee \neg M$	$(L \vee \neg M) \wedge K$	$(L \vee \neg M) \wedge K \wedge (\neg L)$
T	T	T	F	T	T	F
T	T	F	F	T	F	F
T	F	T	F	T	T	F
T	F	F	F	T	F	F
F	T	T	T	F	F	F
F	T	F	T	F	F	F
F	F	T	T	T	T	T
F	F	F	T	T	F	F

Therefore, there is only one possible combination of truth values for K , L , and M .

Namely,

- L is false,
- M is false,
- and K is true.

■

1.2 (b)

Problem 2. Let G , H , and J be statement variables. Use a truth table to prove that

$$H \wedge (G \iff J) \equiv (G \wedge (H \wedge J)) \vee \neg(H \implies (G \vee J)).$$

Solution. Here is the truth table for $H \wedge (G \iff J)$

H	G	J	$G \iff J$	$H \wedge (G \iff J)$
T	T	T	T	T
T	T	F	F	F
T	F	T	F	F
T	F	F	T	T
F	T	T	T	F
F	T	F	F	F
F	F	T	F	F
F	F	F	T	F

Here is the truth table for $(G \wedge (H \wedge J)) \vee \neg(H \implies (G \vee J))$

H	G	J	$H \wedge J$	$G \wedge (H \wedge J)$
T	T	T	T	T
T	T	F	F	F
T	F	T	F	F
T	F	F	F	F
F	T	T	F	F
F	T	F	F	F
F	F	T	F	F
F	F	F	F	F

H	G	J	$G \vee J$	$H \implies (G \vee J)$	$\neg(H \implies (G \vee J))$
T	T	T	T	T	F
T	T	F	T	T	F
T	F	T	T	T	F
T	F	F	F	F	T
F	T	T	T	T	F
F	T	F	T	T	F
F	F	T	T	T	F
F	F	F	F	T	F

H	G	J	$(G \wedge (H \wedge J)) \vee \neg(H \implies (G \vee J))$
T	T	T	T
T	T	F	F
T	F	T	F
T	F	F	T
F	T	T	F
F	T	F	F
F	F	T	F
F	F	F	F

Since $H \wedge (G \iff J)$ and $(G \wedge (H \wedge J)) \vee \neg(H \implies (G \vee J))$ have the same truth table value. Thus, these two statements are equivalent. ■

2 Problem 2

(a):

Problem 3. State the hypothesis of $J(q, r)$

Solution. $q + 2 \neq 0$ or $r^2 - 1 \neq 0$. ■

(b)

Problem 4. State the conclusion of $J(q, r)$

Solution.

$$(q + 2)(r^2 - 1) \neq 0$$

■

(c)

Solution.

$L(q, r) :$ "If $(q + 2)(r^2 - 1) = 0$, then $q + 2 = 0$ and $r^2 - 1 = 0$."

That is false. ■

(d)

Solution.

$K(q, r) : \text{"If } (q + 2)(r^2 - 1) \neq 0, \text{ then } q + 2 \neq 0 \text{ or } r^2 - 1 \neq 0."$

That is true. ■

(e)

Solution.

$$\forall r \in \mathbb{R}, \exists q \in \mathbb{R}, [(q + 2 \neq 0) \vee (r^2 - 1 \neq 0) \implies ((q + 2)(r^2 - 1) \neq 0)]$$

This statement is true. ■

(f): This statement is true.

3 Problem 3

- (a): This is false.

Solution. Suppose, there exists five consecutive integers whose average is equal to the second-largest of those integers. Let n be an arbitrary integer in $\mathbb{Z}$. We set these five integers are

$$n-2, \quad n-1, \quad n, \quad n+1, \quad \text{and} \quad n+2.$$

By the assumption, we have

$$n+1 = \frac{(n-2) + (n-1) + n + (n+1) + (n+2)}{5}.$$

By expanding the numerator, we get

$$n+1 = \frac{n-2+n-1+n+n+1+n+2}{5}.$$

By adding up the numerator, we get

$$n+1 = \frac{5n+0}{5} = \frac{5n}{5}.$$

Simplifying, we get

$$n+1 = n,$$

which is a contradiction to the fact that $n+1 \neq n$.

Therefore, the original claim does not hold. ■

- (b) This is true.

Solution. Suppose that k is an arbitrary integer, and k^2 is the first perfect square of the three consecutive perfect squares.

Therefore, the three perfect squares are

$$k^2, \quad (k+1)^2 \quad (k+2)^2.$$

Take average of these three perfect squares, we have

$$avg = \frac{k^2 + (k+1)^2 + (k+2)^2}{3}.$$

By expanding each term and adding them together, we get

$$avg = \frac{k^2 + k^2 + 2k + 1 + k^2 + 4k + 4}{3} = \frac{3k^2 + 6k + 5}{3}.$$

Then, we can split $3k^2 + 6k + 5$ into $3k^2 + 6k + 3 + 2$, we get

$$avg = \frac{3k^2 + 6k + 3 + 2}{3} = \frac{3k^2 + 6k + 3}{3} + \frac{2}{3} = k^2 + 2k + 1 + \frac{2}{3}.$$

So there exists $c = k^2 + 2k + 1$ in $\mathbb{Z}$ such that

$$avg = c + \frac{2}{3}.$$

Thus, we proved that the average of three consecutive perfect squares is always a non-integer. ■

4 Q4

- (a)

Problem 5. Prove the following statement:

$$\forall x \in \mathbb{R}, \quad 2|5x + 15| - 4|3x - 18| \leq 90.$$

Solution. Let us take a look at the absolute value first.

First, we find under what condition, $5x + 15 \geq 0$.

By subtracting 15 to both sides and dividing 5, we get

$$5x \geq -15$$

$$x \geq -3$$

Therefore, when $x \geq -3$, $5x + 15 \geq 0$.

When $x < -3$, $5x + 15 < 0$.

Next, we find under which condition, $3x - 18 \geq 0$.

By adding 18 to both sides and dividing 3, we get

$$3x \geq 18$$

$$x \geq 6$$

Therefore, when $x \geq 6$, $3x - 18 \geq 0$.

When $x < 6$, $3x - 18 < 0$.

There are three cases to consider:

$$- \quad x < -3$$

Under this condition, both $5x + 15$ and $3x - 18$ are less than 0. So we have

$$2[-(5x + 15)] - 4[-(3x - 18)].$$

Expanding the parentheses, we have

$$2[-5x - 15] - 4[-3x + 18].$$

Expanding the brackets, we have

$$-10x - 30 + 12x - 72.$$

Adding all the terms together, we get

$$2x - 102$$

Since we assume $x < -3$, we get

$$2x - 102 < 2(-3) - 102 = -6 - 102 = -108 \leq 90.$$

Thus, this statement holds true.

$$- \quad -3 \leq x < 6$$

Under this condition, $5x + 15$ is positive, $3x - 18$ is negative.

So we have

$$2|5x + 15| - 4|3x - 18| = 2(5x + 15) - 4[-(3x - 18)]$$

Expanding the bracks, we have

$$10x + 30 - 4[-3x + 18] = 10x + 30 + 12x - 72.$$

Simplifying, we get

$$22x - 42$$

Since $x < 6$ we get

$$22x - 42 < 22(6) - 42 = 90.$$

Therefore, the inequility holds for $-3 \leq x < 6$.

– Case 3: $x \geq 6$.

Under this condition, both $5c + 15$ and $3x - 18$ are positive.

So we have

$$2|5x + 15| - 4|3x - 18| = 2(5x + 15) - 4(3x - 18)$$

Expanding the bracks, we have

$$10x + 30 - 12x + 72$$

Simplifying, we get

$$-2x + 102$$

Since $x \geq 6$, we have

$$-2x + 102 \leq -2(6) + 102 = 90.$$

Since this inequility holds for $x < -3$, $-3 \leq x < 6$, and $x \geq 6$. Therefore, it holds for all x in real number set.

■

5 Q5

Problem 6. Prove the following statement:

$$\forall p, q, r \in \mathbb{R}, \quad 2(p^2 + 13r^2) + q(5q + 4p - 10r) \geq 2pr.$$

Solution. Suppose p, q, r are all arbitrary real numbers in $\mathbb{R}$.

According to the real numbers property, we have

$$(p - r)^2 \geq 0 \quad (p + 2q)^2 \geq 0 \quad (q - 5r)^2 \geq 0,$$

Adding all the inequilities together, we get

$$(p - r)^2 + (p + 2q)^2 + (q - 5r)^2 \geq 0.$$

Following from the square of binomial,

$$(p^2 - 2pr + r^2) + (p^2 + 4pq + 4q^2) + (q^2 - 10qr + 25r^2) \geq 0.$$

Expanding the parenthesis, we have

$$p^2 - 2pr + r^2 + p^2 + 4pq + 4q^2 + q^2 - 10qr + 25r^2 \geq 0.$$

Simplifying the inequility, this yields:

$$2p^2 + 26r^2 + 5q^2 + 4pq - 10qr - 2pr \geq 0.$$

Adding $2pr$ for both sides, we have

$$2p^2 + 26r^2 + 5q^2 + 4pq - 10qr \geq 2pr.$$

Factoring 2 out of $2p^2 + 26r^2$ and q out of $5q^2 + 4pq - 10qr$, we get

$$2(p^2 + 13r^2) + q(5q + 4p - 10r) \geq 2pr.$$

Therefore, the original inequility is true. ■

6 Q6

- (a) Open Sentence 1: $S(x)$
Open Sentence 2: $U(x)$
Open Sentence 3: $M(x)$

- (b) Open Sentence 4: $P(x)$
- Open Sentence 5: $Q(x)$
- Open Sentence 6: $T(x)$

(c)

$$\exists x \in \mathbb{R}, (x^2 - 36 = 0) \wedge (x^2 - 10x + 24 \neq 0) \wedge (x^2 - 16 \neq 0)$$

(d) Let's say. Pick -6 to be $x \in \mathbb{R}$, we have

$$x^2 - 36 = (-6)^2 - 36 = 36 - 36 = 0$$

and

$$x^2 - 10x + 24 = (-6)^2 - 10(-6) + 24 = 120 \neq 0$$

and

$$x^2 - 16 = (-6)^2 - 16 = 20 \neq 0$$

Therefore, there exists x in $\mathbb{R}$ such that $(x^2 - 36 = 0) \wedge (x^2 - 10x + 24 \neq 0) \wedge (x^2 - 16 \neq 0)$.